



AI Chatbot using PocketFlow & Google Gemini



Project Overview

This project is a command-line AI chatbot built using **PocketFlow** and **Google Gemini AI**. The chatbot accepts user input, sends the conversation to the Gemini Large Language Model (LLM), and generates intelligent, context-aware responses. It maintains conversation history, enabling natural and interactive chats.

The project demonstrates how workflow orchestration with PocketFlow can be combined with Google's Gemini model to build a simple conversational AI assistant.



Features

- Interactive command-line chatbot
 - Powered by Google Gemini AI
 - Maintains conversation history
 - Workflow orchestration using PocketFlow
 - Clean and modular Python code
 - Secure API key configuration
 - Type `exit` to end the conversation
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Technologies Used

- Python 3.x
 - PocketFlow
 - Google Gemini API
 - Google Generative AI SDK
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Project Structure

```
AI-Chatbot/  
|— chatbot.py      # Main chatbot application  
|— README.md      # Project documentation  
|— requirements.txt # Required Python packages
```

Installation

1. Clone the repository

```
git clone https://github.com/your-username/AI-Chatbot.git  
cd AI-Chatbot
```

2. Create a virtual environment (Optional)

```
python -m venv .venv
```

Activate it:

Windows

```
.venv\Scripts\activate
```

Linux/macOS

```
source .venv/bin/activate
```

3. Install dependencies

```
pip install pocketflow  
pip install google-generativeai
```

Or install all dependencies:

```
pip install -r requirements.txt
```

Configure Gemini API

Create a Google Gemini API key from Google AI Studio.

Replace the placeholder in the code:

```
API_KEY = "YOUR_GEMINI_API_KEY"
```

with your own API key.

Run the Project

```
python chatbot.py
```

Example:

```
Welcome to AI Chatbot  
Type 'exit' to quit.
```

```
You: Hello
```

```
Assistant: Hello! How can I help you today?
```

```
You: What is Artificial Intelligence?
```

```
Assistant: Artificial Intelligence (AI) is the simulation of human intelligence  
by machines...
```

Workflow

1. User enters a message.
 2. PocketFlow receives the input.
 3. The conversation history is updated.
 4. Gemini AI processes the request.
 5. The chatbot displays the response.
 6. The process repeats until the user types `exit`.
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Future Enhancements

- Streamlit web interface
- Voice-enabled chatbot
- File upload and document chat
- PDF Question Answering
- Retrieval-Augmented Generation (RAG)
- Multi-language support
- Database-backed chat history
- Authentication and user profiles

- REST API integration
 - Deployment on cloud platforms
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Demo

Add screenshots or a GIF of your chatbot here.

Learning Outcomes

This project helped in understanding:

- Workflow orchestration with PocketFlow
 - Large Language Models (LLMs)
 - Google Gemini API integration
 - Conversation memory
 - Prompt engineering
 - Python application development
 - AI chatbot architecture
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Contributions

Contributions are welcome! Feel free to fork the repository, create a new branch, and submit a pull request.

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Author

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